UNIT 3 — CLOUD SECURITY

Practical Report: Set Up a Web Application on Amazon EC2

Purpose: launch a Linux EC2 instance, configure network access securely, install a web server, deploy HTML, and verify the site.

Important screenshot note

The interface images in this report are realistic-style **illustrative mockups**, not captures from a live AWS account or an actual Windows VM. Replace them with your own screenshots before submitting if your instructor requires evidence of your completed lab.

Prepared as a step-by-step practical guide

1. Objective

Deploy a basic HTML web application on an Amazon EC2 Linux instance, allow secure administration over SSH, and verify the site over HTTP.

2. Launch an EC2 instance

• Sign in to the AWS Management Console and open EC2 in the intended region.

• Choose Launch instance. Select Amazon Linux 2023, choose a lab-appropriate instance type, and create/select a key pair.

• Choose a VPC/subnet that can provide internet access. If the website must be public, the instance needs a public IPv4 address and a route to an internet gateway.

• Create a security group with SSH TCP 22 sourced only from My IP (/32) and HTTP TCP 80 from 0.0.0.0/0. For IPv6 public access, add an appropriate IPv6 HTTP rule only if needed.

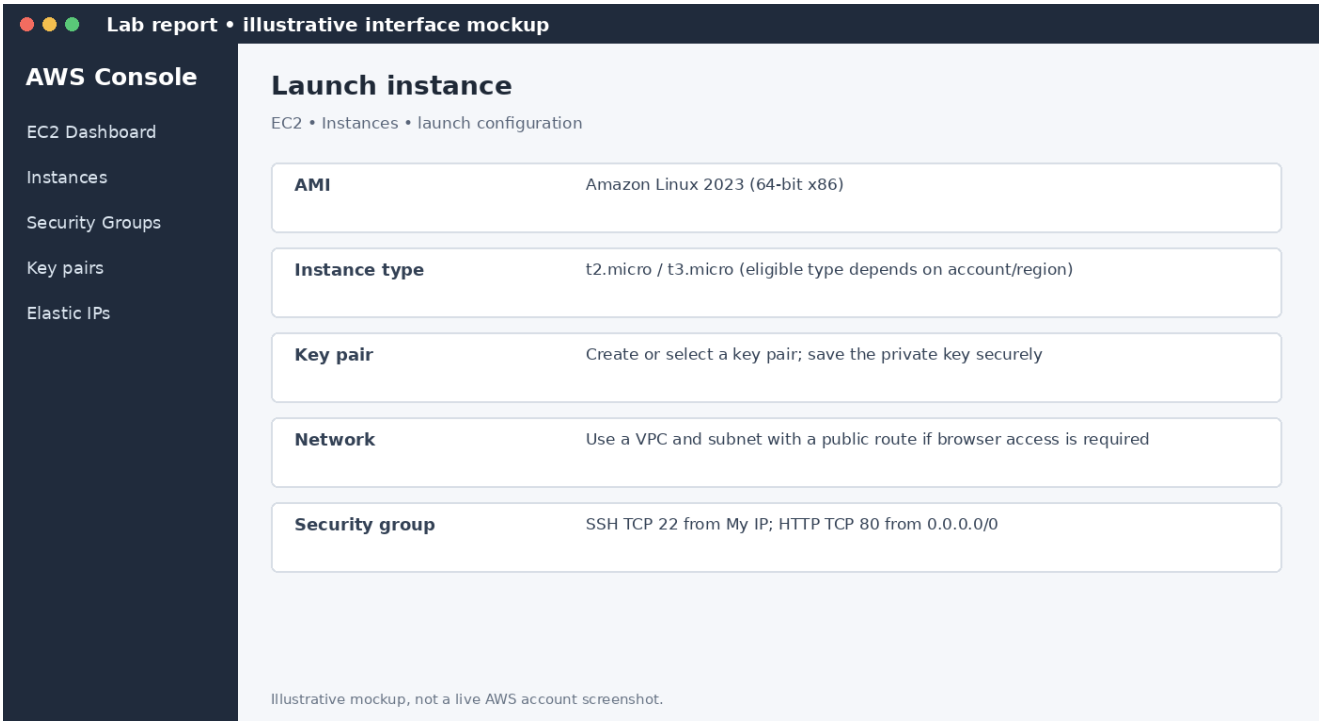


Figure 1. Example EC2 launch settings. Names and values are illustrative; verify the options in your account.

3. Configure and verify the security group

In EC2 → Security Groups → Inbound rules, confirm the two required rules below. Restrict SSH to your current public IP address. HTTP is public only because the sample website is intended to be reachable in a browser.

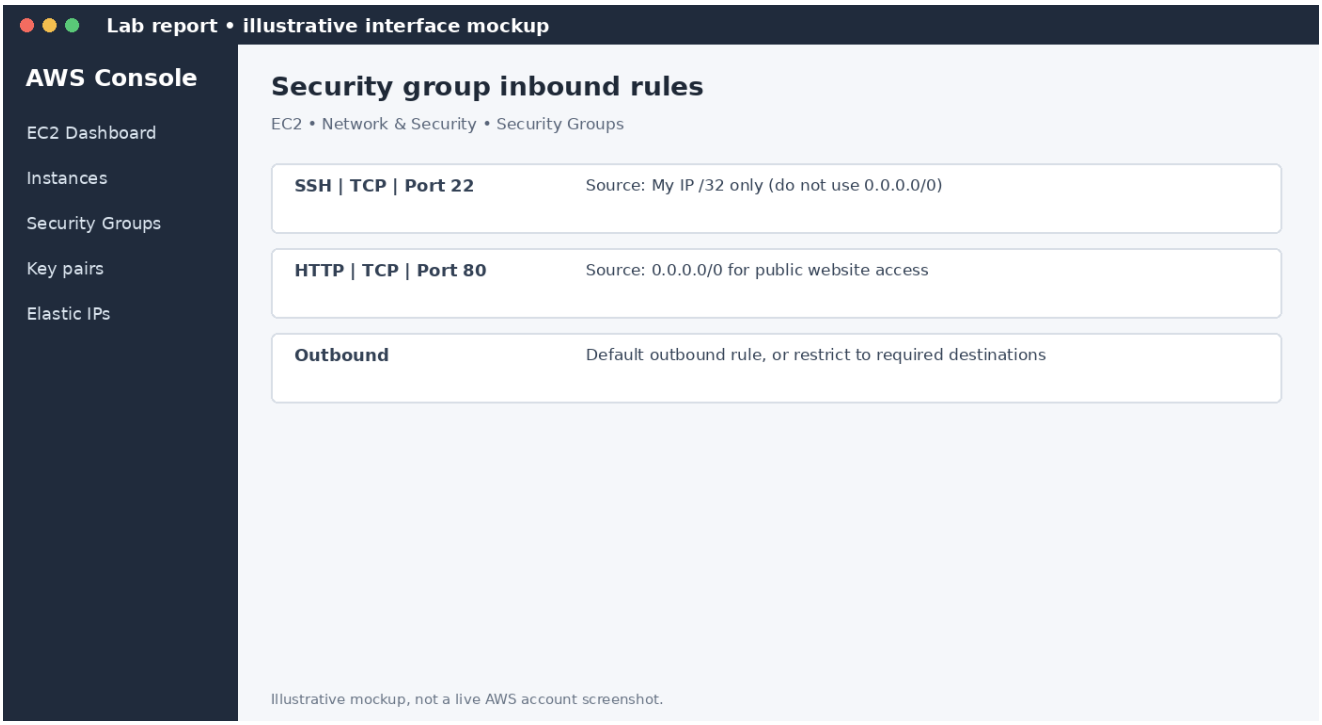


Figure 2. Example inbound-rule layout. Set the SSH source to your actual IP; do not copy a placeholder.

4. Connect using SSH

On Linux/macOS, use a terminal. On Windows, use PowerShell or an SSH client. Replace the file name and public IP with your real values.

```
chmod 400 your-key.pem
ssh -i your-key.pem ec2-user@YOUR_PUBLIC_IP
```

If the connection fails, verify the instance is running, the public IP is correct, the subnet route allows internet access, and TCP 22 is allowed from your current IP. Do not share your private key.

5. Install Apache and deploy HTML

```
sudo dnf update -y
sudo dnf install -y httpd
sudo systemctl enable --now httpd
sudo systemctl status httpd --no-pager
echo '<h1>Hello from Amazon EC2</h1><p>Unit 3 practical</p>' | sudo tee
/var/www/html/index.html
```

If your chosen operating system uses a different package manager, follow the equivalent installation commands for that OS. Ensure the web server service is active.

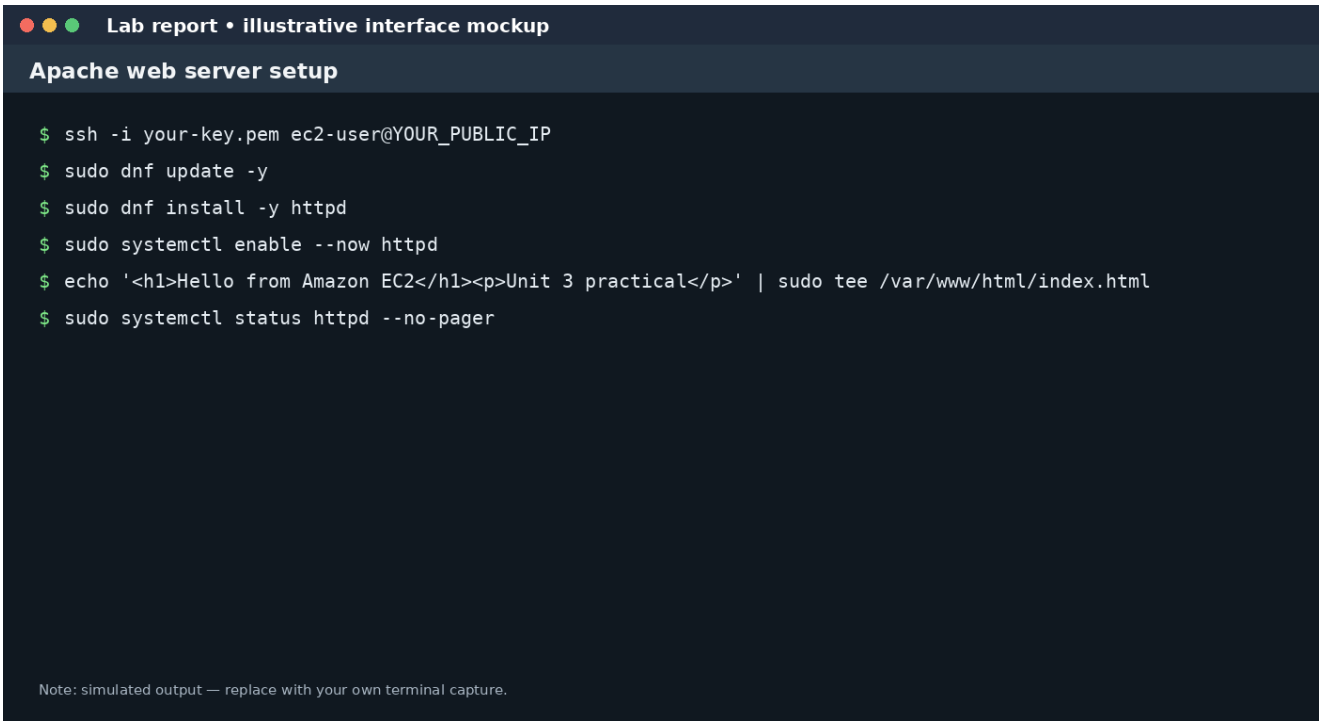


Figure 3. Example terminal commands and service check. Output is simulated; capture your actual terminal for submission.

6. Verify in a browser

Open http://YOUR_PUBLIC_IP in a browser, replacing the placeholder with your instance's public IPv4 address. The page should display the HTML message. If it does not, check the Apache service, security group HTTP rule, subnet route, and any OS firewall settings.

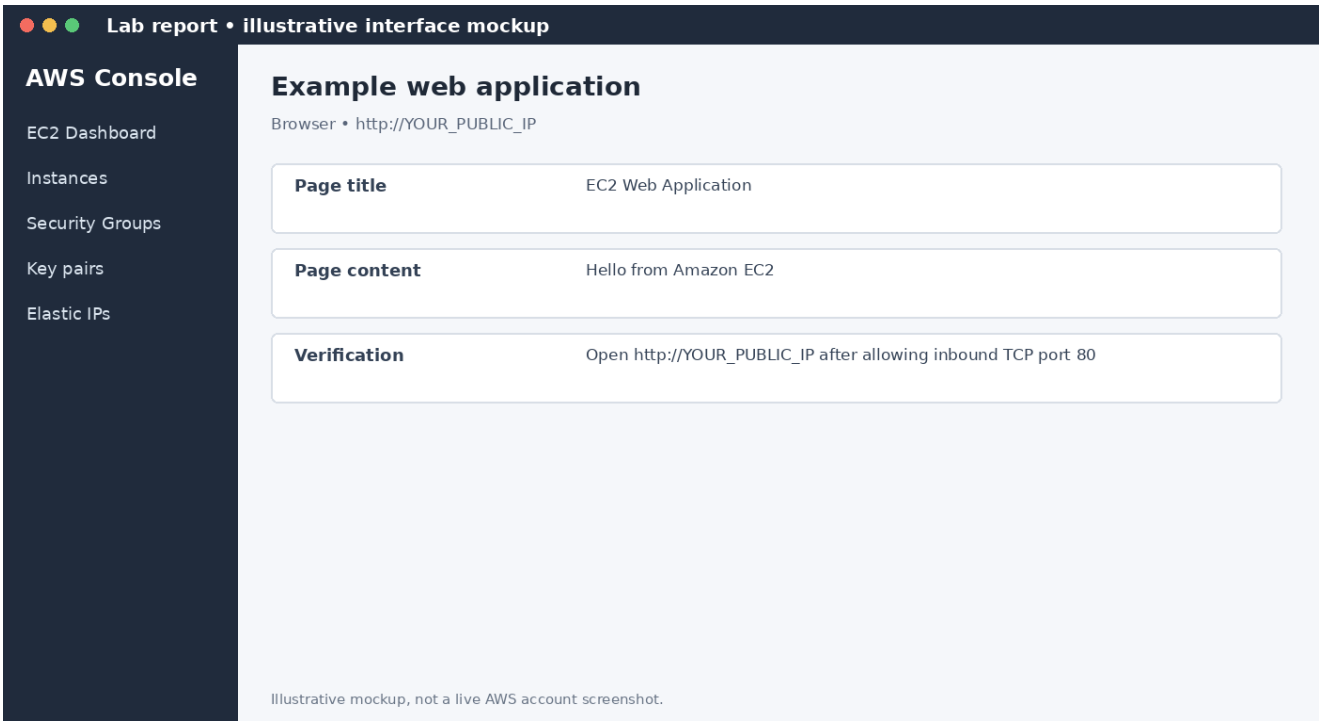


Figure 4. Example expected browser result. This is a mockup, not a live endpoint.

7. Security and cleanup

• Keep SSH limited to My IP (/32), and never publish the private key.

• Do not put credentials or private keys in screenshots or the report.

• After taking your real screenshots, stop or terminate the instance when the assignment is complete to avoid unnecessary charges. Termination is irreversible.

8. Screenshot checklist

Capture your own evidence for: (1) EC2 instance details with public IP partially masked if desired, (2) inbound security group rules, (3) successful SSH session and Apache status, and (4) the web page opened using the instance IP. Avoid exposing account IDs, access keys, or private key material.

Conclusion

The practical is complete when the EC2 instance is reachable by SSH from the permitted IP, Apache is running, and the HTML page loads over HTTP. Record the actual instance details and replace all illustrative screenshots with your own evidence.